



# Ecological networks of an Antarctic ecosystem: a full description of non-trophic interactions

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## Abstract

Interactions between organisms are diverse and attend to multiple biological demands; hence, understanding ecological communities requires considering different types of species interactions beyond predation. In this work, we assembled the non-trophic networks of a marine Antarctic ecosystem for the first time. We report mutualistic (+/+), competitive (−/−), commensalistic (+/0) and amensalistic (−/0) interactions between species of the Potter Cove marine community (South Shetland Is., Antarctica). Based on a network approach, we present a full description of each type of interaction and analyze its distribution according to different species-level properties. Also, we constructed a multiple interactions network including trophic and non-trophic interactions and studied network-level properties. We found more than double non-trophic interactions than trophic mostly corresponding to competitive interactions that mainly involve mid-trophic level species. Low-trophic level species were mainly involved in mutualistic and amensalistic interactions. We observed that single-type interaction networks display differences in their topology. Finally, we highlight that including a description of species interactions in ecological network analyses provides a better understanding of ecosystems which it is crucial to comprehend and predict ecosystems responses to environmental changes.

**Keywords** Non-trophic interactions · Ecological networks · Species-level properties · Network-level properties · Multiple interactions' network

## Introduction

Trophic interactions among species and food web topological analysis led to multiple applications in ecological communities and many scientific advances have been developed (Paine 1966; Cohen 1977; Post and Pimm 1983; Cohen and Briand 1984; McCann et al. 1998; Martinez et al. 1999; Williams and Martinez 2000; Dunne et al. 2002; Netuel et al. 2002; Pascual and Dunne 2006; Albouy et al. 2014, among many others). Nevertheless, it is widely known that

interactions between co-occurring species of an ecological community involve much more than just predator–prey interactions (Kéfi et al. 2012, 2015, 2016; Pocock et al. 2012; Mougi 2016a). Indeed, many studies have reported different types of relationships between species and, consequently, different types of non-trophic interactions in a given community (Bloom 1975; Kneib 1991; Amsler et al. 1999; McClintock et al. 2005; Bascompte and Jordano 2008). A particular example of this occurs in the West Antarctic Peninsula (WAP) where many species that are directly (or indirectly) related have been reported for mutual benefits such as macroalgae species and amphipods through chemical mediation (Aumack et al. 2010; Amsler and McClintock 2014) and epibiotic interactions (Dayton et al. 1974; Gutt 1998, 2000).

Non-trophic interactions have been studied in many communities considering different approaches. For example, Hacker and Gaines (1997) presented a qualitative theoretical model that considers how direct positive interactions (mutualism and commensalism) affect community species diversity. Mougi and Kondoh (2012) showed that multiple interaction types are key to understanding community

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dynamics and suggested that antagonistic (competitive) and mutualistic interactions might stabilize population dynamics. Also, Mougi (2016a, b) showed that non-trophic interactions such as amensalistic, commensalistic and mutualistic play a crucial role in communities' persistence. Non-trophic interactions have also been studied from a network perspective, considering different types of interactions among species. For instance, Bascompte and Jordano (2007, 2008, 2009) studied the structure and complexity of mutualistic networks, suggesting that these networks can be regarded as the architecture of biodiversity. Furthermore, Bastolla et al. (2009) showed that the number of co-existing species in a community can be determined by both the structure of mutualistic networks and that of competition for common resources. In the last decade, some studies have analyzed trophic and non-trophic networks to understand the patterns and dynamics of diverse species interactions in nature (Kéfi et al. 2012, 2015, 2018; Pockock et al. 2012; Miele et al. 2019). Some works focused on ecosystems' stability by developing theoretical (Thébault and Fontaine 2010) or analytical approaches (Allesina and Tang 2012), while others have focused on dynamic modeling (Kéfi et al. 2016; Miele et al. 2019). More recently, a few studies have incorporated multiple interactions networks integrating more than two interaction types in a single network (Kéfi et al. 2016; Pilosof et al. 2017; García Callejas et al. 2018; Hervías-Parejo et al. 2020). It is noteworthy that studies on species interactions in Antarctic ecosystems have focused mainly on analyzing trophic interaction networks neglecting other types of interaction (Marina et al. 2018a; Cordone et al. 2020; McCormack et al. 2021).

Recently, the study of marine Antarctic ecosystems has increased rapidly and the South Shetland Islands have been the principal study area for many investigations (Knap et al. 1996; Hall and Perry 2004; Hass et al. 2010; Simms et al. 2012; Pellizzari et al. 2017; Burfeid-Castellanos et al. 2021, among others). Potter Cove is an Antarctic fjord located in 25 de Mayo/King George Island (62°14'S, 58°40'W) which belongs to the South Shetland Islands. This Cove has been described as a spatially heterogeneous spot regarding its bottom characteristics: the hard-substrate outer cove colonized by macroalgae and the soft-sediments inner cove with high concentration of filter feeders (Klöser et al. 1994; Quartino et al. 2005; Cordone et al. 2020). Furthermore, Potter Cove represents one of the most biodiverse fjords of the WAP (Tatian et al. 2004; Grange and Smith 2013). It is not only a biodiversity hotspot but also one of the marine ecosystems where drastic environmental and ecological changes are happening due to climate change (Schloss et al. 2012; Quartino et al. 2013; Sahade et al. 2015; Hernández et al. 2019). In this sense, an accurate assessment of the structure and complexity of Potter Cove community might be crucial to understand how other WAP fjord ecosystems could respond

to climate change (Vaughan et al. 2003; Turner et al. 2005; Meredith and King 2005; Bromwich et al. 2013; Nicolas and Bromwich 2014). In this regard, Potter Cove food web has been recently described and analyzed applying a network perspective (Marina et al. 2018a; Cordone et al. 2020), but little is known about the structure of non-trophic ecological networks in Potter Cove. Incorporating non-trophic interactions could add a new perspective and yield unexpected results about species role and about how the ecosystem might respond to environmental changes (Kéfi et al. 2012; Mougi 2016a, b).

In this work, we present and characterize four networks of non-trophic interactions among species from the Potter Cove marine ecosystem and describe the distribution of the interactions in relation to species properties. We: (1) provide a detailed description of each type of interactions among species, (2) analyze the distribution of the non-trophic interactions, and (3) assemble trophic and non-trophic networks in a highly resolved multiple interactions network.

## Methods

### Species interactions

#### Non-trophic interactions

The identification of mutualistic (+/+), commensalistic (+/0) and amensalistic (−/0) interactions was done by compiling information through an extensive bibliographic search. We considered publications related to the set of biological species and trophic species (aggregated taxa) described by Marina et al. (2018a) when presenting the food web of the Potter Cove marine ecosystem. From now on in the text, we will use the term “species” to refer to both biological and trophic species. The bibliography selection process was made only considering published articles that provide information about non-trophic interactions between species from the Potter Cove food web (see Supplementary Material).

Competitive interactions (−/−) were established, on one hand, from the predator secondary graph, the so-called competition network. This network was obtained from the food web (or primary graph) of Potter Cove described in Marina et al. (2018a) and a competitive interaction was considered when two predators shared at least one prey. On the other hand, we included competition for other resources—such light and space—among benthic species of the macroalgae community and invertebrate sessile organisms (e.g., Porifera). These interactions were added considering reported data of benthic species from Potter Cove. We included more than 80 articles where information of non-trophic interactions was provided (Supplementary Material—Appendix I).

Finally, we assembled four networks considering the following types of interactions: mutualistic, commensalistic, amensalistic and competitive. Most of the species considered were identified at species level (e.g., *Euphausia superba*), but some were identified as a group of biological species (e.g., Phytoplankton). Every network was plotted with the software Visone 2.20.

### Trophic interactions

Information on co-occurring species and their feeding habits was obtained mostly from publications resulting from a cooperation program between Argentina and Germany that started in 1994 and continued for more than 20 years (Wiencke et al. 1998, 2008). Based on these data, Marina et al. (2018a) presented the Potter Cove food web including 91 species and 307 interactions. Here, we considered the same set of species and incorporated such trophic relationships ( $\pm$ ) for constructing the multiple interactions network described below.

### Network assemblage

#### Non-trophic networks

Mutualistic (+/+), commensalistic (+/0) and amensalistic (-/0) networks were first described as binary matrices representing presence (1) or absence (0) of an interaction. Competitive network was based on the definition of a secondary graph for predators sharing at least one prey (Box I). Based on this graph, we constructed an interaction list where we added competition for other resources such as light and space.

#### Multiple interactions' network

We included five types of interactions among the 91 species from Potter Cove: trophic (+/-), mutualistic (+/+), commensalistic (+/0), amensalistic (-/0) and competitive (-/-).

Network construction was developed considering a square matrix,  $M$ , where each interaction was represented by a pair of integers in the interval  $[-1, 1]$ . The sign of the non-zero elements indicates if that species benefits (+) or not (-) from the interaction. For example, a competitive interaction between nodes A and C is represented by the pair  $(-1; -1)$  in the  $M_{AC}$  and  $M_{CA}$  positions (Fig. 1). A commensalistic interaction is represented by the pair  $(1; 0)$  or  $(0; 1)$  depending on which species benefits from that interaction. Analogously, an amensalistic interaction is represented by the pair  $(0; -1)$  or  $(-1; 0)$ . Trophic interactions were represented considering which species is the prey and which is the predator in the feeding relationship, i.e.,  $(1; -1)$  or  $(-1; 1)$ ,

|   | A  | B | C  | D  | E  |
|---|----|---|----|----|----|
| A | 0  | 1 | -1 | 0  | 1  |
| B | 0  | 0 | 0  | 0  | -1 |
| C | -1 | 0 | 0  | 0  | 0  |
| D | 0  | 0 | 0  | 0  | 0  |
| E | 1  | 1 | 0  | -1 | 0  |

**Fig. 1** Representation of the different types of interactions in the square matrix associated with the multiple interactions network. Each interaction was identified by a combination of integers between -1 and 1

where -1 is the prey and 1 is the predator. Finally, mutualism has a single representation pair in the matrix  $(1; 1)$ , since both species benefit from that interaction.

### Network analysis

We quantified a set of network-level properties to describe the structure and complexity of each studied network: (1) number of species ( $S$ ); (2) total number of interactions or links ( $L$ ), (3) density ( $L/S$ ), and (4) percentage of basal ( $B$ , without prey), intermediate ( $I$ , with prey and predator), and top ( $T$ , without predator) species.

We considered connectance ( $C$ ) as the ratio between observed ( $L$ ) and possible interactions ( $S^2$ )

$$C = L/S^2, \quad (3)$$

where  $L$  is the total number of interactions or links and  $S$  is the number of species in the interaction matrix. It is noteworthy that this equation includes the possibility of self-links among species. We have decided to follow a conservative approach and use such equation, rather than  $C = L/S(S-1)$  that neglects self-links, since some of the species of the networks are not taxonomically resolved at the biological species level, causing the appearance of self-links in networks that a priori should not have them, such as competitive self-links among Ascidiae and Bryozoa species. This property is considered an estimator of community sensitivity to perturbations and it covaries with other network properties (Dunne et al. 2002; Fortuna et al. 2008; Poisot and Gravel 2014).

We also explored species-level properties with the aim of identifying those species that are more important considering the total number and the distribution of its (trophic and non-trophic) interactions. For this, we analyzed: (1) species degree as the sum of incoming and outgoing interactions

for each species (all prey and predators of a species in the trophic network, for instance); (2) basal, intermediate, and top categories; and (3) trophic level for each species. Basal species are those with predators but without prey, intermediate species have both prey and predators, and top species have prey but no predators. The trophic level for each species was calculated as one plus the mean trophic level of all of the species' resources, where the trophic level of a resource is the chain length from the resource to a basal species (Williams and Martinez 2004). Properties (2) and (3) were included here based on the trophic network data provided by Marina et al. 2018a. To know if the distribution of non-trophic interactions in Potter Cove differed from the expected by chance, we compared the observed distribution of non-trophic interactions among basal, intermediate, and top species with those obtained from 1000 randomized networks considering the same number of species (S), links (L), and percentage of basal (B), intermediate (I), and top (T) species than Potter Cove. This process was developed using the software MatLab v2020.

### Box 1

We described non-trophic networks as complex networks defined by  $n \times m$  binary matrices with the form

$$A = [\alpha_{ij}]^{n \times m}. \quad (1)$$

The matrix  $A$  describes the interactions between the sets of species  $n$  and  $m$  and its elements,  $\alpha$ , represent the presence or absence of an interaction in the web  $F$  as follows:

$$\alpha_{ij} = \begin{cases} 1, & k_{ij} \in (F) \\ 0, & k_{ij} \notin (F) \end{cases}, \quad (2)$$

where  $k_i, k_j$  are two any nodes of the  $n$  and  $m$  set, respectively, and  $k_{ij} \in F$  indicates the interaction between  $i$  and

$j$ . Note that if  $n=m$ ,  $A$  would be an adjacency matrix that could represent, for example, a food web.

The construction of the competitive (for food) network was developed based on the definition of secondary graph and can be explained as follows:

Given the food web  $F$ , the vertices of the competition graph called  $G(F)$  are the same as those of  $F$ , i.e., one vertex for each species in the community. The edges of  $G(F)$  between distinct vertices  $i$  and  $j$  are undirected and represent an overlap between diets of species  $i$  and  $j$ , i.e., these edges exist if and only if there exists some third vertex  $k$  in  $F$ , such that  $i$  eats  $k$  and  $j$  eats  $k$ . Thus, in  $G(F)$ , two vertices are linked by an edge if there are arrows in  $F$  from  $k$  to  $i$  and from  $k$  to  $j$ , for at least one  $k$ ; or if one row  $k$  of the adjacency matrix has elements equal to 1 in both column  $i$  and column  $j$  (Cohen 1977).

## Results

### Non-trophic networks

In the Potter Cove marine ecosystem, there were more than twice non-trophic interactions than trophic. Within non-trophic interactions, the majority were among species competing for the same prey, space, and light (competitive = 80%), followed by mutualistic (12%), commensalistic (4.1%) and amensalistic (3.9%) (Table 1).

More than 60% of the mutualistic interactions identified corresponded to relationships between herbivorous amphipods (mesograzers) and macroalgae species. Most of commensalistic and amensalistic interactions (> 90%) corresponded to epibiotic relationships, i.e., relationships between two different organisms in which one of them serves as a substrate for the other, e.g., ascidians and/or epiphytic diatoms on macroalgae. We considered the latter as an amensalistic interaction (−/0), because epiphytes compete with macroalgae for light and nutrients (Amsler and McClintock 2014) and epiphytes do not especially need the algae

**Table 1** Complexity and structural properties of the non-trophic networks

|                               | S  | L    | L/S   | C    | B   | I  | T  |
|-------------------------------|----|------|-------|------|-----|----|----|
| Mutualistic (+/+)             | 36 | 88   | 2.44  | 0.07 | 67  | 33 | 0  |
| Commensalistic (+/0)          | 23 | 29   | 1.26  | 0.06 | 57  | 35 | 9  |
| Amensalistic (−/0)            | 25 | 24   | 0.96  | 0.04 | 100 | 0  | 0  |
| Competitive (−/−)             | 74 | 566  | 7.65  | 0.10 | 20  | 58 | 22 |
| Trophic (+/−)                 | 91 | 307  | 3.37  | 0.04 | 34  | 47 | 19 |
| Multiple interactions network | 91 | 1014 | 11.14 | 0.12 | 34  | 47 | 19 |

S number of species, L number of interactions, L/S linkage density, C connectance. B, I, and T percentage of basal, intermediate, and top species respectively

to survive, since they can grow on a different substrate type such as rocks or sediments. Detailed information about species interactions is provided in Supplementary Appendix II.

### Network-level properties

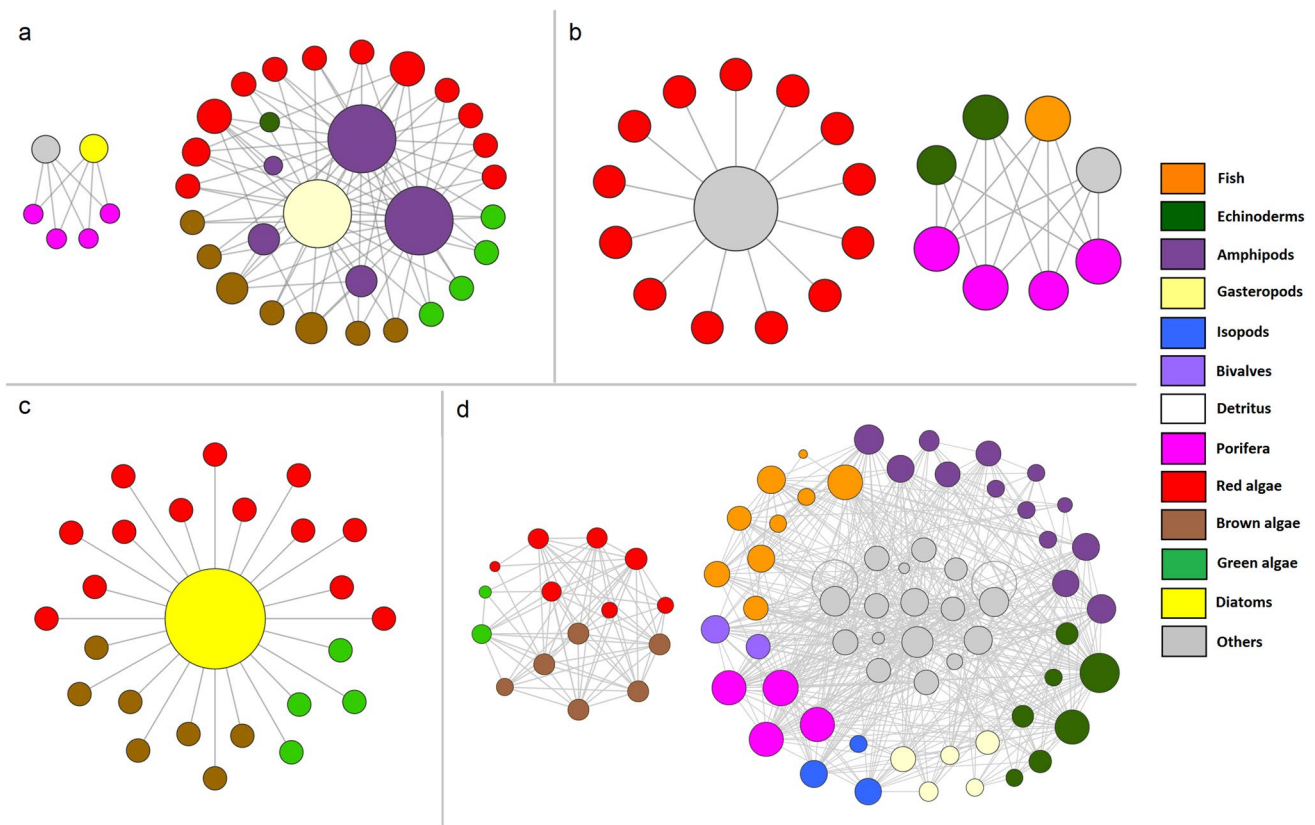
Regarding network-level properties, most non-trophic networks displayed a relatively low number of species (S) and interactions (L). However, connectance (C) and linkage density (L/S) were relatively higher in non-trophic networks than in the trophic network (food web); competitive (−/−) and mutualistic (+/+) networks presented the highest values for L/S. The multiple interactions network presented the highest value for L/S and C. Furthermore, non-trophic networks showed distinctive characteristics considering basal/intermediate/top species: (1) mutualistic and amensalistic networks did not involve top species and (2) in the competitive network, interactions occurred mostly among intermediate species (Table 1).

The topology of the non-trophic networks differed depending on the type of interaction: mutualistic, commensalistic, and competitive networks exhibited a fragmented topology (Fig. 2a, b, d), while the amensalistic network

displayed a single component with only one species connected to the others (Fig. 2c). Finally, the multiple interactions network displayed a hyper-connected web with the highest L/S and C (Fig. 3, Table 1).

### Species-level properties

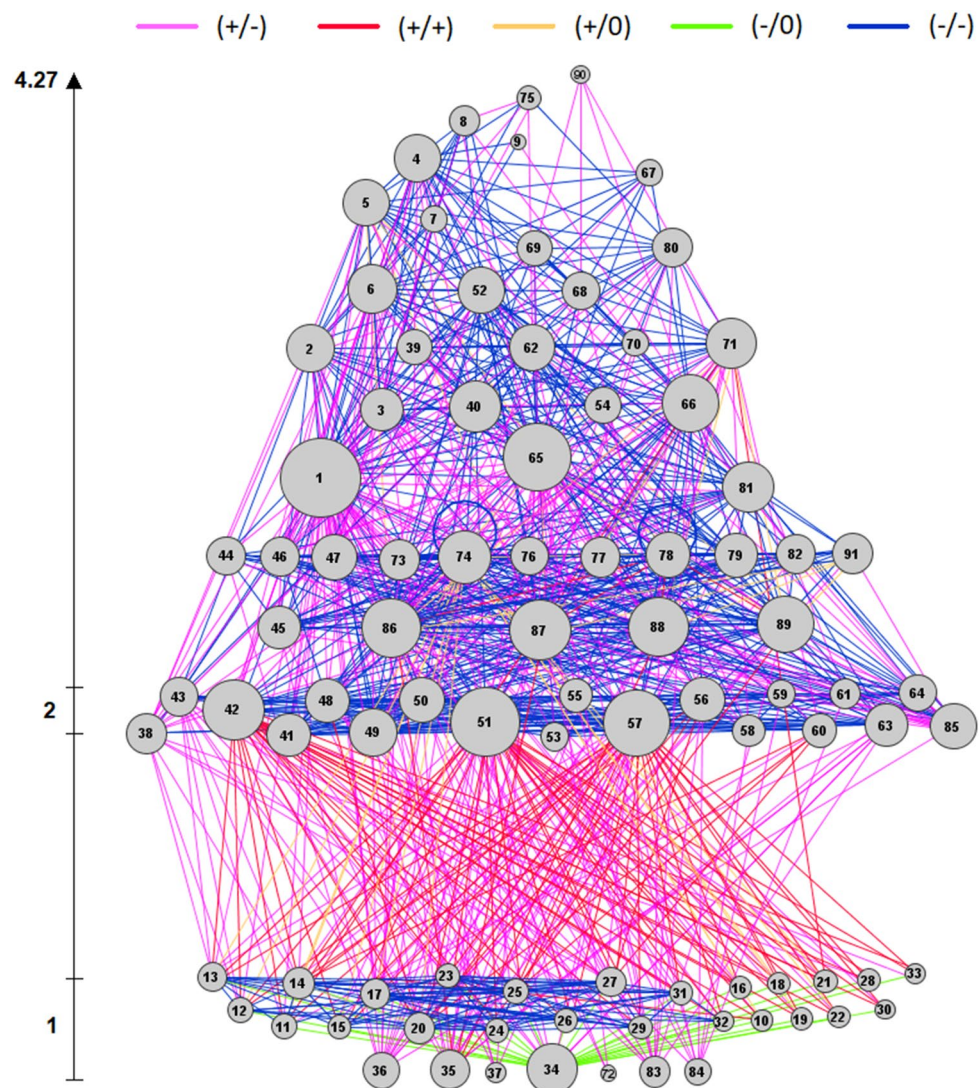
When we explored species-level properties, we observed that species of the Potter Cove marine ecosystem with the highest number of non-trophic interactions were also at mid-trophic levels (intermediate species) (Fig. 4, 5). Only one species with the most (upper 10%) non-trophic interactions was a top species (*Notothenia coriiceps*, trophic level = 2.80). This upper 10% is distributed between species of a variety of functional groups, all of them closely related to the benthos (Echinodermata, Amphipoda, Porifera, Gastropoda, demersal Fish). For most of these species, the dominant type of non-trophic interaction was competition followed by mutualism (Fig. 4, 5). On the other hand, the species with the least (bottom 10%) non-trophic interactions were macroalgae species, where mutualism ruled their relationships, and invertebrate mobile species where competition for prey dominated their interactions (e.g., octopus) (Fig. 4).



**Fig. 2** Graphic representation of non-trophic networks: **a** mutualistic, **b** commensalistic, **c** amensalistic and **d** competitive. Each circle (node) represents a species in the network. Functional groups are

indicated with different colors. The category “others” corresponds to single-species groups (e.g., Zooplankton)

**Fig. 3** Multiple interactions network of the Potter Cove marine ecosystem. Vertical position indicates the trophic level of each species (node). The size of the circles (nodes) is proportional to its degree (total number of interactions) and the number corresponds to species ID (Supplementary Appendix III)

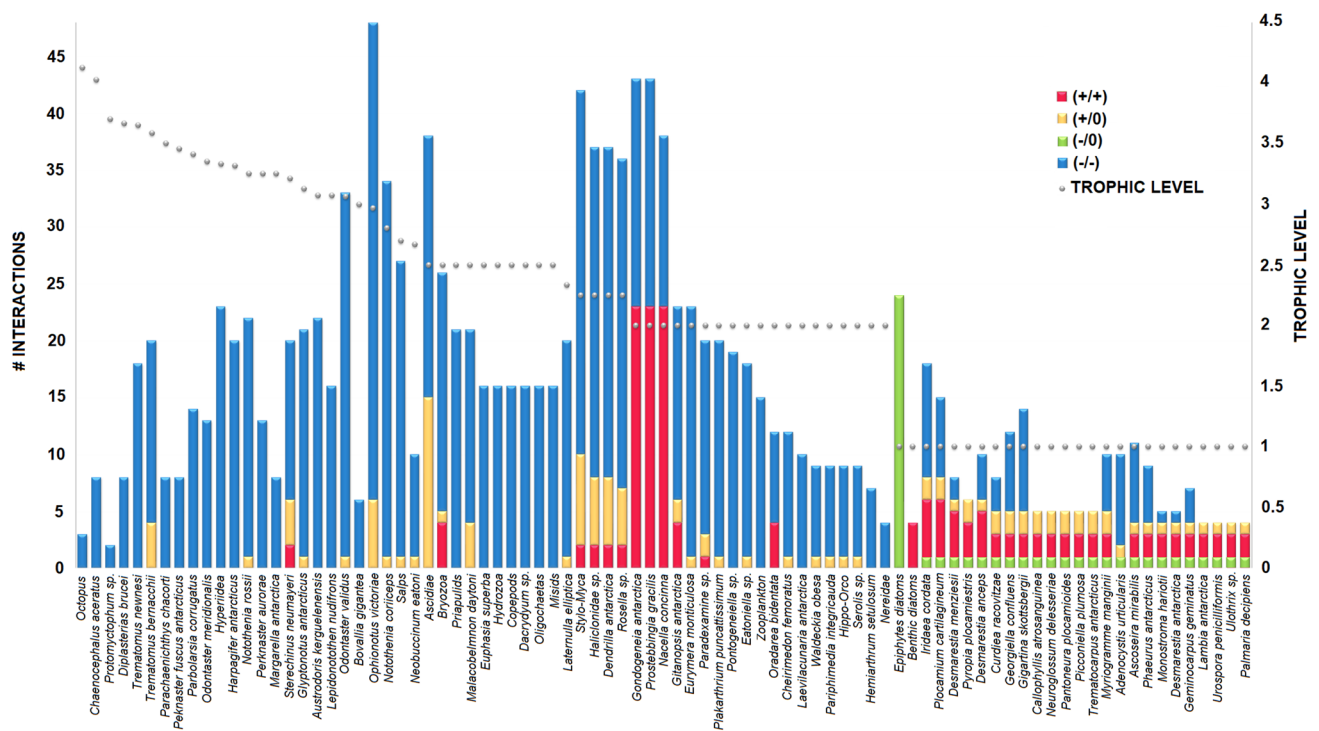


Top species, without predators, exhibited a wide range in the number of non-trophic interactions (min = 8, max = 34). The great majority of these interactions were for competing for the same prey (Fig. 2d, 5) which represented 85% of the total competitive interactions. Detailed information about species interactions and trophic levels is provided in Supplementary Appendix II and III, respectively.

As we mentioned, the majority (62.4%) of non-trophic interactions in Potter Cove were distributed among intermediate species and ruled by competitive interactions. Since the percentage of intermediate species in Potter Cove represents almost 50% of the richness, we explored if there were more non-trophic interactions among intermediate species than expected by chance (Supplementary Appendix III). We found that the number of non-trophic interactions among intermediate species in Potter Cove was significantly higher

than expected by chance. We also observed that non-trophic interactions among basal species were significantly lower than expected by chance (Fig. 5b).

It is important to note that species from all trophic levels presented commensalistic interactions, although in relatively low numbers. On the contrary, amensalistic interactions occurred in species at low-trophic levels (basal species) (Fig. 5). Some of the species with the highest number of this type of interaction (e.g., epiphytic diatoms) were the ones with the highest number of non-trophic interactions within basal species (Fig. 4). Considering all interaction types (multiple interactions network), the species with the highest number of interactions were also those with the highest number of non-trophic interactions; at mid-trophic levels, as mentioned above (Fig. 3).



**Fig. 4** Number of non-trophic interactions per species. Vertical axes indicate total number of interactions (left) and trophic level (right). Interactions: mutualistic (+/+), commensalistic (+/0), amensalistic (-/0) and competitive (-/-)

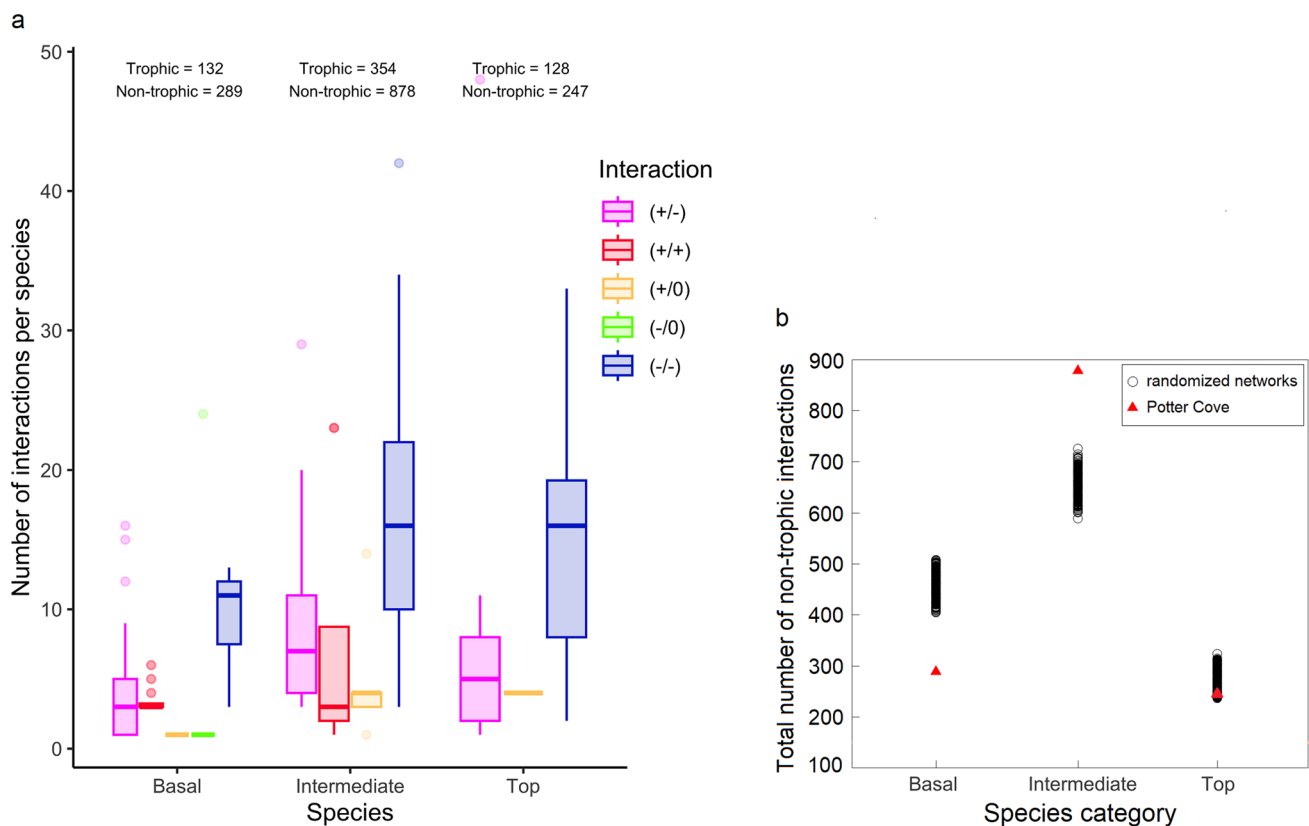
## Discussion

This work presents the first description of non-trophic interactions for an Antarctic ecosystem including also trophic interactions in a highly resolved network. The network-level properties of the multiple interactions network were different from those of the single-interaction networks. This work provides an overview of the species interactions in the community according to different properties and feeding strategies.

### Distribution of interactions

The distribution of the non-trophic interactions identified in Potter Cove was mostly represented between intermediate trophic level species. This result might not be related to the fact that the percentage of intermediate species in Potter Cove food web is close to 50%, that it is not as high as other marine food webs (Dunne et al. 2004; Vermaat et al. 2009; Marina et al. 2018b). A large number of trophic interactions at this trophic level are related to a large number of competitive interactions; a large number of competitive interactions are, ultimately, related to the criterion used here to define competition for food (sharing at least one prey). That is, the over-representation of competitive interactions is a consequence of the excessive dietary overlapping among intermediate species in Potter Cove food web. In this sense,

amphipods and demersal fish were the trophic guilds that dominated competitive interactions. Also, competitive interactions included competition for light and space (dominated by macroalgae and invertebrate sessile organisms), though these interactions were represented in a relatively low number (15%) compared with competitive interactions for food, because they do not imply feeding relationships. On the other hand, the non-trophic/non-competitive interactions were mainly associated with invertebrate sessile (e.g., Porifera) and basal species that provide habitat, shelter or facilitate feeding for other species (e.g., macroalgae species). For example, mutualistic interactions involve mesograzers that are often found in close association with macroalgae. The macroalgae are chemically defended from being eaten by mesograzers and also provide them a chemically defended refuge from predation by higher species level (omnivorous fish). Then, the macroalgae is benefited, since mesograzers feed on epiphytic algae that can compete with the macroalgae for light and nutrients (Aumack et al. 2010; Amsler and McClintock 2014). Furthermore, commensalistic and amensalistic interactions involved filtering species that take advantage of the elevated position (in relation to the sea floor) that provides better access to food due to the higher speed of currents (Gutt 2000).



**Fig. 5** a Distribution of trophic and non-trophic interactions per species considering basal (without prey), intermediate (with prey and predator) and top (without predator) species. Interactions: trophic (+/-), mutualistic (+/+), commensalistic (+/0), amensalistic (-/0) and competitive (-/-). In this boxplot, the middle “box” represents the middle 50% of values (interquartile range IQR), the horizontal line within each box is the median, the vertical lines are the lower (Q1-1.5\*IQR) and upper (Q3+1.5\*IQR) quartiles, and points show

outliers. b) Comparison of total number of non-trophic interactions per species category (basal, intermediate and top) between Potter Cove and random networks. Black circles segments correspond to the 95% confidence intervals of 1000 random networks generated with the same number of species (S), links (L) and percentage of basal (B), intermediate (I) and top (T) species than Potter Cove, represented with red triangles. The simulation of the random networks was developed with the software MatLab v2020

## Species network properties

In Potter Cove marine ecosystem, the classification of species into basal, intermediate and top species, their diet, their trophic level and their mobility (mobile or sessile) are important elements related to the number of non-trophic interactions in which species participate. For example, species at relatively higher-trophic levels (intermediate and top species) mainly participate in competitive interactions and most invertebrate sessile species (e.g., filter feeders) participate in commensalistic interactions. In this sense, it might be possible to extrapolate these patterns to other polar benthic communities (Pineda-Metz et al. 2019; Bae et al. 2021).

Some studies argue that studying both trophic and non-trophic networks allows us to better understand community complexity and that the combination of different structural patterns of interactions networks is essential to comprehend, for example, the mechanisms behind communities' stability (Mougi and Kondoh 2012; Kéfi et al. 2016; García-Callejas

et al. 2018; Freilich 2018). In this regard, previous analyses of Potter Cove food web showed that there is an important energetic subsidy from detritus and basal species to the entire community and suggested that topological features such as the proportion of basal, intermediate and top species are key to understand the ecosystem response against diversity loss (Marina et al. 2018a; Cordone et al. 2018, 2020). Hence, the importance of incorporating different types of species interactions (trophic and non-trophic) at different trophic levels in a network framework. More specifically, adding non-trophic interactions and their distribution among species might be fundamental to understanding particular topological properties of the ecosystem (Marina et al. 2018a, 2018b). In addition, the network approach might be necessary to build realistic and reliable ecological models (Kéfi et al. 2015, 2016; Mougi 2016a, 2016b) that allow, for example, to disentangle those species with important functional implications in the ecosystem (Sanders et al 2015).

## Non-trophic interactions' role

The role of non-trophic interactions in ecosystems might be ambiguous and probably dependent on many factors such as the type of community being studied (terrestrial, aquatic, etc.) or the analysis performed (Kéfi et al. 2012). Regardless of this, it is clear that non-trophic interactions are key elements that should be considered when studying ecological communities' dynamics and stability (Kéfi et al. 2012, 2015, 2016, 2018; Mougi 2016b; García-Callejas et al. 2018; Hervías-Parejo et al. 2020). There were some attempts to address this and the findings were contradictory between model predictions and natural ecosystems. On the one hand, Allesina and Tang (2012) suggested that trophic interactions increase stability, while mutualistic and competitive interactions destabilize it. Nevertheless, real ecosystems such as the Chilean rocky shores, presenting numerous positive and negative interactions, showed a persistent and resilient response to perturbations (Kéfi et al. 2015, 2016). In this sense, the following question should be addressed: if mutualistic and competitive interactions destabilize ecosystems, why are they so commonly found in ecological communities as empirical data suggest? One possible answer could be that analytical models might not be sufficient when analyzing aspects of real ecosystems. This could be, partly, because mathematical reduction assumes an abstraction with the consequent loss of attributes and characteristics of the ecological system. However, mathematical modeling of ecosystems allows us to tackle life's complex phenomena (Pascual and Dunne 2006). Then, increasing the realism of such models, as in the present study by incorporating non-trophic interactions, will help us to better understand the structure and dynamics of ecosystems (Mougi et al. 2012; Miele et al. 2019; Ward et al. 2022).

Studies performed by Kéfi et al. (2015, 2016) are the only ones that presented the non-trophic interactions of a well-known marine ecosystem from a network perspective, as far as we know. These authors explored the ecological networks of a Chilean rocky intertidal community considering trophic and non-trophic (negative and positive) interactions and were able to build a three-dimensional (three different interactions) multiplex network. They also assessed the role of non-trophic interactions on the ecosystem function using a dynamical modeling approach considering species traits to describe the complexity of the community. One of their findings is related to the high number of non-trophic interactions (dominated by competitive interactions as in Potter Cove) among basal species because of the disproportionate basal species richness (Kéfi et al. 2015). This result differs from what was found in the present work, where most of the non-trophic interactions in Potter Cove are among intermediate species, which is not related with the high

richness of intermediate species but with the criterion used here to define competition for food. We identified a competitive interaction if two predators shared at least one prey. If a more restrictive criterion is chosen, for example that predators share more than 50% of the prey, the number of competitive interactions would be less. We selected this criterion, since it is the original definition of the secondary graph (Cohen 1977). Another difference is related to the level of description of the non-trophic interactions, since Kéfi et al. (2015, 2016) did not describe all possible type of non-trophic interactions among species but considered three sets of interactions: positive (e.g., habitat/refuge provisioning by sessile species that create structure for others), negative (e.g., interference, competition for space) and trophic interactions. The present work is pioneer in the description of five types of non-trophic interactions between species of an Antarctic ecosystem.

A few ecosystems have been studied considering how non-trophic interactions are distributed among species or if the amount of these interactions is related with some topological and structural features of the community; even the association between species-level properties and interaction types remains unclear. In this sense, including a full description of the species and its relationships in ecological network analyses—like the one presented here—is of matter importance and provides a realistic insight into the structure and complexity of ecosystems (Kéfi et al. 2012, 2015, 2016; Moughi 2016a, 2016b; Lurgi et al. 2020). Therefore, addressing species properties analysis according to the type of interaction is necessary, because it might shed light on how to better assess and predict ecosystems responses to environmental disturbances, especially when it comes to ecosystems already being stressed by climate change, such as the Antarctic ones (Sahade et al. 2015; Hernández et al. 2019).

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**Data availability statement** The datasets generated and analyzed during the current study are available in the Figshare repository: <https://doi.org/10.6084/m9.figshare.21369855>. Files included: Spreadsheets (Appendices I, II and III) + Decision tree (PDF).

## Declarations

**Conflicts of interest** The authors have no relevant financial or non-financial interests to disclose.

**Ethics approval** This study did not require ethical approval, since it only uses data from species studied and published in academic articles.

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